ExamForge™ System Architecture

Zero-Trust, Anti-Collusion Examination Infrastructure & Enterprise EaaS Platform

Target Audience: Technical Architects, C-Suite, Examination Boards	**Release Version:** v2.0 Enterprise Release
Security Model: Cryptographic Zero-Trust & Shamir Secret Sharing	**Date:** August 2026

1. Executive Overview & Industry Challenge

Modern high-stakes national and institutional examinations face unprecedented vulnerabilities: physical question paper leaks, impersonation at test centers, candidate collusion through seating manipulations, catastrophic operational batch errors, and black-box evaluation opacity. ExamForge was engineered from the ground up as a **Zero-Trust Examination Infrastructure** that eliminates single points of trust across every lifecycle touchpoint.

Industry Vulnerability	ExamForge Zero-Trust Architecture Solution
1. Paper Leaks & Premature Access: Vulnerable custody storage and unencrypted paper delivery.	**Dual-Custody Key Ceremony:** Split keys via *(m-of-n)* Shamir Secret Sharing. Papers unseal only at T-15 min with supervisor quorum.
2. Impersonation & Ghost Candidates: Manual gate check-in vulnerable to forged admit cards.	**Offline UIDAI Biometric Verification:** Cryptographic QR signatures and facial biometric matching without cloud latency.
3. Spatial Collusion & Seating Fraud: Predictable alphabetical seating allows neighboring peer cheating.	**Anti-Collusion Graph Seating:** Graph-theoretic spatial dispersion ensuring zero cohort proximity.
4. Catastrophic Bulk Errors: Dangerous all-or-nothing operations with no blast-radius previews.	**ExamForge SafeBatch:** Impact preview, resilient exception isolation, and automated operational handoffs.
5. Opaque Grading & Disputes: Black-box score modifications without public verifiability.	**Merkle Tree Forensic Ledger:** Mathematically provable, tamper-evident audit receipts for all events.

2. Multi-Tiered System Architecture

ExamForge employs a high-throughput, horizontally scalable 4-tier architecture with strictly defined boundaries, cryptographic telemetry tracing, and tenancy isolation.

Architectural Layer	Components & Technologies	Primary Responsibilities

Tier 1: Presentation & Portals	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS, Lucide Glass UI, PWA Service Worker	10 dedicated persona workspaces, sub-50ms glass interactions, offline CBT exam renderer, real-time telemetry widgets.
Tier 2: API Gateway & Middleware	FastAPI, Pydantic v2, TenantMiddleware, RequestIDMiddleware, SecurityHeadersMiddleware	Multi-tenant context isolation, cryptographic trace correlation, RBAC enforcement across 25+ domain routers, rate-limiting.
Tier 3: Core Domain & Crypto Engines	Key Ceremony Engine, Anti-Collusion Graph, SafeBatch Studio, Dual-Blind Evaluator, AI Blueprinting, Merkle Prover	Secret sharing orchestration, pre-flight blast radius simulation, spatial seating optimization, automated score conflict arbitration.
Tier 4: Persistence & Event Stream	PostgreSQL / SQLite, SQLAlchemy ORM (40+ models), Encrypted S3 Artifact Store, WebSockets / SSE	Authoritative state storage, tamper-evident Merkle hash tree links, biometric archive, live incident event stream.

3. Role-Based Workspaces & Persona Matrix

ExamForge completely eliminates shared generic dashboards by provisioning dedicated, role-isolated command centers.

Workspace / Persona	Target Route	Key Functional Capabilities
1. Authority & Controller	`/authority, /controller`	Exam blueprinting, question difficulty balance, dual-key unsealing ceremony, publication gates.
2. Vendor Partner (EaaS)	`/vendor`	Third-party agency onboarding, hardware verification, SLA monitoring, bulk candidate allocations.
3. Centre Superintendent	`/center-console`	Center desk, physical seat map locking, gate biometrics, buffer capacity override, handoff claims.
4. Candidate / Student	`/candidate, /student-exam`	UIDAI registration, admit card QR, distraction-free CBT console, cryptographic score breakdown.
5. Subject Evaluator	`/evaluator, /omr-review`	Dual-blind grading queue, rubric scoring, OMR computer vision scanner, conflict arbitration queue.
6. Independent Auditor	`/audit-timeline`	Cryptographic Merkle tree timeline, root-hash inspector, immutable dispute version ledger.
7. War Room Commander	`/war-room`	Real-time node connectivity radar, active incident kill-switch, live center packet synchronization.
8. Security & Pen-Test Lead	`/security-pentest`	Automated red-team simulated attacks, key rotation lifecycle, zero-trust policy compliance auditor.
9. Dispute Appeals Officer	`/disputes, /dispute-ops`	Candidate grievance triaging, signed evidence packet generator, referee re-evaluation tracking.
10. SafeBatch Operations Hub	`/safebatch`	Safeguarded bulk allocation wizard, pre-flight blast simulation, operational handoff management.

4. End-to-End Examination Lifecycle

ExamForge governs the entire assessment pipeline across 5 secure chronological phases:

- **Phase 1: Pre-Examination & Blueprinting:** AI topic weighting, Bloom's taxonomy distribution, question authoring with dual-custody paper encryption and (m-of-n) secret key generation.
- **Phase 2: Secure Allocation & Packaging:** ExamForge SafeBatch bulk allocation, encrypted offline center package distribution (.efpkg), and anti-collusion spatial seating generation.
- **Phase 3: Exam Execution & Gate Verification:** Offline UIDAI QR biometric gate check-in, real-time War Room telemetry, distraction-free offline-resilient CBT exam taking.
- **Phase 4: Evaluation & Arbitration:** OMR computer vision extraction, dual-blind subjective grading, and automated referee conflict arbitration when evaluator score delta exceeds 2.0.
- **Phase 5: Results, Merkle Proofs & Certificates:** Scorecard Merkle tree sealing, ECDSA tamper-evident QR certificates, public cryptographic proof verification portal.

5. ExamForge SafeBatch: Safeguarded Bulk Operations

High-volume operations (e.g., allocating 2,847 candidates across centers) traditionally risk irreversible data corruption or ambiguous failures. ExamForge SafeBatch introduces **blast-radius impact previews, resilient exception isolation, and automated operational handoffs**.

Step	SafeBatch Stage	Mechanism & Output
01	**Scope & Risk Classification**	Evaluates operation risk level (Low, Medium, High, Critical) and selects candidate cohort.
02	**Pre-Flight Impact Preview**	Simulates allocation across 4 centres. Identifies 2,813 safe allocations, 23 capacity overruns, and 11 missing address flaws before touching DB.
03	**Safety Confirmation Gate**	Requires explicit operator sign-off and dual-custody authorization for high/critical tier operations.
04	**Resilient Safe Execution**	Allocates 2,813 safe candidates immediately. Isolates the 34 exceptions into discrete exception records without batch rollback.
05	**Operational Handoff Note**	Generates structured handoff packet (HO-2026-0822-0034) assigned to Centre Superintendent with 1-click buffer seat resolution console.

6. Cryptographic Innovation & Zero-Trust Protocols

ExamForge implements mathematical guarantees rather than relying on administrative trust:

- **Shamir's (m, n) Threshold Secret Sharing:** Examination papers are encrypted with AES-256-GCM. Unsealing requires a quorum of cryptographic keys held by independent officials, preventing single-actor leaks.
- **Spatial Anti-Collusion Seating Algorithm:** Constructs candidate adjacency graphs factoring in roll numbers, institutions, and postal codes to mathematically ensure zero proximity between affiliated test-takers.
- **Offline Public-Key Biometric Gate Check:** Parses UIDAI QR codes using local RSA-2048/ECDSA public certificate chains, enabling instant identity validation even in total network blackout conditions.

- **Merkle Tree Forensic Audit Ledger:** Every state mutation emits a SHA-256 hash linked into the Merkle root. Independent auditors can verify the mathematical proof of any exam event.

7. Technical Stack & Production Topology

Stack Layer	Selected Technology	Key Rationale
Frontend Framework	Next.js 16 (App Router), React 19, TypeScript	Server Components, edge rendering, zero hydration drift, sub-second load times.
Styling & Design	Tailwind CSS, Lucide Icons, Glassmorphism	High-contrast pine green/mint aesthetic, fully responsive across tablets and desktops.
Backend API Core	Python 3.12/3.14, FastAPI, Pydantic v2	High-concurrency async endpoints, automated OpenAPI contract generation, strict typing.
Database & ORM	SQLAlchemy ORM, PostgreSQL / SQLite	40+ relational models, automated schema migration, ACID transactional consistency.
Real-Time Telemetry	WebSockets, Server-Sent Events (SSE)	Sub-100ms node latency monitoring, live War Room radar, batch execution progress.
Edge Deployment	Vercel Edge, Docker Containerized Nodes	Global CDN caching, zero-config deployment, local center node containerization.

8. Presentation Slide Flow (PPT Guide)

When delivering this architecture to stakeholders, structure the presentation into 8 core slides:

- **Slide 1: Vision & Problem:** The 5 critical vulnerabilities in legacy assessments and ExamForge's zero-trust thesis.
- **Slide 2: 4-Tier Architecture:** Visual diagram showing Presentation, API Gateway, Crypto Engines, and Persistence.
- **Slide 3: 10 Persona Workspaces:** Role-based command centers for Controllers, Vendors, Superintendents, and Evaluators.
- **Slide 4: 5-Phase Exam Lifecycle:** End-to-end journey from AI blueprinting to Merkle score publishing.
- **Slide 5: SafeBatch Operations:** Blast-radius simulation, partial execution, and automated operational handoff notes.
- **Slide 6: Cryptographic Innovations:** Dual-custody key ceremony, offline UIDAI QR check, and anti-collusion seating graph.
- **Slide 7: Evaluation & Dispute Engine:** Dual-blind grading, OMR computer vision scanner, and automated arbitration queue.
- **Slide 8: Enterprise Tech Stack & ROI:** Next.js 16 + FastAPI + PostgreSQL topology and competitive advantages.